MIL-C-46808(MI)
29 MARCH 1963
SUPERSEDING
MPD-1256
15 MAY 1958

MILITARY SPECIFICATION

CONTROL, FREQUENCY, AUTOMATIC 8151460

1. SCOPE

1.1 This specification covers one type of control, designated as Control, Frequency, Automatic, 8151460.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-152 —Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.

MIL-T-945 —Test-Equipment, for use with Electronic Equipment: General Specification.

MIL-E-5272 —Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-130 —Identification Marking of U.S. Military Property.

MIL-STD-202 —Test Methods for Electronic and Electrical Component Parts.

MIL-STD-252 —Wired Equipment, Classification of Visual and Mechanical Defects for.

DRAWINGS

ORDNANCE CORPS

8151460 —Control, Frequency, Automatic (Reference Unit).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules

(Application for copies should be addressed to: Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Preproduction sample. When a preproduc-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:59Z
Check the source to verify that this is the current version before use.

tion sample is required (see 6.2), it shall meet all the requirements of this specification. The preproduction sample shall be manufactured using the same methods proposed for the manufacture of subsequent production lots.

3.2 Materials. Materials used in construction of each unit shall be new, and in accordance with Drawing 8151460 and applicable drawings and specifications referenced thereon.

3.3 Design and construction. The design and construction of the unit shall be in accordance with Drawing 8151460 and applicable drawings and specifications referenced thereon.

3.4 Electrical characteristics.

3.4.1 *Ground potential.* The design and construction of each unit shall be such that all external metallic parts are at ground potential at all times.

3.4.2 *Circuitry resistance.* The circuitry resistance, between each combination of terminals of P1 and from each terminal of P1 to circuit ground shall be in accordance with Drawing 8151460 and applicable drawings and specifications referenced thereon.

3.4.3 *Insulation resistance.* Unless otherwise specified on applicable drawings, all insulation shall have a minimum resistance of 100 megohms between isolated circuits and between ungrounded circuits and ground.

3.5 Performance characteristics.

3.5.1 *Input voltage.* The unit shall operate satisfactorily when supplied with input voltage as specified in table I.

TABLE I. *Power required*

Voltage	Current	Frequency	Phase	Remarks
6.3	3.7 amp	400 cps	1	±0.3 volts
+150	.050 amp	dc		±5.0 volts
+350	.030 amp	dc		±10.0 volts
−250	.001 amp	dc		±5.0 volts

3.5.1.1 *Power consumption.* The power consumption shall be such that the current shall not exceed that specified in table I.

3.5.2 *Search cycle time.* With no input to J1, the search cycle period of the unit under test shall be 20 seconds plus or minus 1 second.

3.5.3 *Search voltage.* The voltage measured between J3 and ground of the unit under test shall be equal to or more negative than minus 35 volts when the dim electrode of I1 is illuminated and a positive 16 volts plus or minus 1.5 volt when the bright electrode is illuminated.

3.5.4 *1kc Oscillator output.* The output at P1–11 of the unit under test shall be between 2 and 5 volts. The frequency shall fall between 1,000 cps and 1,400 cps.

3.5.5 *Phase inverter voltage level.*

3.5.5.1 *Voltage level at V3 pin 5.* The voltage indicated at V3 pin 5 of the unit under test shall be between 7 and 12 volts, when the adjustment procedures of 4.8.2.6.1 are followed.

3.5.5.2 *Voltage level at V4 pin 5.* The voltage at V4 pin 5 of the unit under test shall be between 7 and 12 volts when the adjustment procedures in 4.8.2.6.2 are followed.

3.5.6 *Sensitivity, zero degrees phase.*

3.5.6.1 *Sensitivity at V4 pin 2.* There shall be sharp positive pulses at V4 pin 2 indicating conduction. Input at J1 shall be less than 10 millivolts (mv) when adjusted in accordance with 4.8.2.7.1.

3.5.6.2 *AFC output J3.* With an input at J1 of 45 millivolts (mv) there shall be no pulses at V3 pin 2, indicating no conduction. There shall be no square wave present at J1. A positive voltage shall be measured when tested in accordance with 4.8.2.7.2.

3.5.7 *Sensitivity 180° phase.*

3.5.7.1 *Sensitivity at V3 pin 2.* There shall be sharp positive pulses at V3 pin 2 indicating con-

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:59Z
Check the source to verify that this is the current version before use.

duction. Input at J1 shall be less than 10 millivolts when adjusted in accordance with 4.8.2.8.1.

3.5.7.2 *AFC output at J3.* With an input at J1 of 45 mv, there shall be no pulses at V4 pin 2, indicating no conduction. There shall be no square wave present at J3. A negative voltage shall be present when tested in accordance with 4.8.2.8.2 and 4.8.2.8.3.

3.6 Environmental.

3.6.1 *Temperature.*

3.6.1.1 *Operating.* The unit shall operate satisfactorily within ambient temperatures ranging from minus 40 to plus 71° Centigrade (C.) (minus 40 to plus 158.8° Fahrenheit (F.)).

3.6.1.2 *Nonoperating.* The unit shall be capable of being stored and transported without damage or impairment of performance in ambient temperatures ranging from minus 53.8 to plus 85° C. (Minus 65 to plus 185° F.).

3.6.2 *Vibration.* The unit shall not be damaged nor shall its performance be impaired when subjected to vibrating frequencies in the range of 10 to 55 cps, total excursion of 0.0156 inch plus or minus 10 percent.

3.6.3 *Shock.* The unit shall be capable of withstanding, without damage, the shocks that occur in normal operating and nonoperating conditions.

3.6.4 *Relative humidity.* The unit shall not be damaged nor shall its performance be impaired when subjected to a relative humidity of 95 percent.

3.6.5 *Fungus resistance.* The unit shall be treated in accordance with Specification MIL-T-152. The unit shall not be damaged nor shall its performance be impaired when subjected to the fungus test specified in 4.8.3.5.

3.6.6 *Salt-laden air.* The unit shall not be damaged nor shall its performance be impaired by salt-laden air.

3.6.7 *Altitude.* The unit shall operate satisfactorily at any altitude from sea level to 10,000 feet above sea level. The unit nonoperating, shall be capable of being transported by air, without damage, at altitudes up to and including 30,000 feet.

3.7 Details of components. Components of the unit shall conform to Drawing 8151460 and applicable drawings and specifications referenced thereon.

3.8 Dimensions. Dimensions and tolerances of the unit shall be in accordance with Drawing 8151460 and applicable drawings referenced thereon.

3.9 Weight. The weight of the unit shall be not greater than 4 pounds.

3.10 Color and finish. Color and finish of the unit shall be in accordance with Drawing 8151460, and applicable drawings referenced thereon.

3.11 Product marking. Product marking of each unit shall be in accordance with Drawing 8151460, and applicable drawings referenced thereon, and Standard MIL-STD-130.

3.12 Workmanship. All components shall be fitted properly and firmly into their respective positions. All mechanical joints and electrical connections shall be mechanically secure and electrically sound. All controls shall operate freely without slipping, sticking, or binding. Electrical connections shall be flexible between shock mounted components and their foundations. Surfaces of glazed ceramic components shall be smooth, uniform and free from cracks. There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely effect its use or suitability for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Govern-

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:59Z
Check the source to verify that this is the current version before use.

ment. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified, (see 6.2), the Government will perform the examinations and test at the contractor's plant. If the preproduction sample does not meet all requirements of this specification, it will be rejected and returned to the contractor. Subsequent units or lots will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Inspection. Unless otherwise specified (see 6.2), inspection for examination and testing shall be 100 percent of production (see 4.8).

4.4 Lot size. Lot size shall not exceed that quantity of Automatic Frequency Controls produced in one continuous operation by the same manufacturer and manufactured using the same drawing, drawing revision, specification, and specification revision.

4.5 Examination. Visual and tactile examination for defects shall be conducted for the purpose of determining compliance with the requirements as specified in 3.2, 3.3, 3.7, 3.8, 3.9, 3.10, 3.11 and 3.12. Standard MIL-STD-252 shall be used when applicable in this examination.

4.6 Test equipment. The test equipment shall meet the minimum requirements of the equipment listed below and shall be used as indicated in performing the test. In the test procedures, the equipment is referred to by item number. The equipment may be mounted separately or in any combination (such as consoles, panels and multipurpose instruments).

Test equipment required

Item	Description
1	Voltmeter, vacuum tube, with input impedance of 10 megohms or higher to measure plus 20 to minus 75 volts dc with an accuracy of plus or minus 3 percent.
2	Oscilloscope—I-245 or equal.
3	Oscillator to supply 1,000 to 1,400 cycles per second at 2 volts rms across 1,000 ohms.
4	Multimeter—TS-352B/U or equal.
5	Audio Isolation Amplifier capable of converting a 1 volt rms, 1,000 to 1,400 cycles per second input signal into two outputs: One with zero° phase shift controllable from 5 to 45 mv, the other with 180° phase shift controllable from zero to 45 mv.
6	Termination, 75 ohms plus or minus 5 percent to mate with Part Number 8581071.
7	Timer for measuring the search cycle period; timer shall measure a minimum of 30 seconds.
8	Power supply, positive 320 volts dc, .030 amp, plus or minus 10 volts.
9	Power supply, negative 250 volts dc, .001 amp, plus or minus 5 volts.
10	Power supply, 6.3 volts ac, 3.7 amps, 400 cps plus or minus 0.3 volts, single phase.
11	Power supply, positive 150 volts dc, .050 amps plus or minus 5 volts.

4.7 Test setup. The test setup shall be as specified in 4.8 through 4.8.2.8.3, Drawing 8151460 and applicable drawings referenced thereon.

4.7.1 *Test methods and procedures.* The following test shall be performed on 100 percent of production with the exception of the power consumption and environmental test which shall be performed on the preproduction sample only. The test specified in 4.8.1 shall be performed prior to the application of power. The test specified in 4.8.2 shall be performed in the sequence indicated.

4.8 Electrical.

4.8.1 *Ground potential.* All external metallic parts shall be measured to ground with item 4 to determine compliance with 3.4.1.

4.8.1.1 *Circuitry resistance.* A circuit resistance check of the unit under test shall be made, using item 4, to determine compliance to the requirement specified in 3.4.2.

4.8.1.2 *Insulation resistance.* The unit under test shall be given the insulation resistance test

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:59Z
Check the source to verify that this is the current version before use.

in accordance with Standard MIL-STD-202, method 302, condition B, and shall conform to the requirements specified in 3.4.3.

4.8.2 *Performance test.*

4.8.2.1 *Input voltage.* The input voltage shall be in accordance with table I, and schematic Drawing 8151462. Item 4 shall be used to measure voltage, in compliance with 3.5.1.

4.8.2.2 *Power consumption.* The power consumption shall be in accordance with table I. Item 4 shall be used to measure the current in compliance with 3.5.1.1.

4.8.2.3 *Search cycle time.* Adjust all controls to full CCW. With no input to J1, the sweep rate adjust control R57 shall be adjusted to obtain a total search cycle of 20 seconds plus or minus one second. Observe the cycling of the bright electrode on indicator lamp I1, measure the period of the search cycle using item 7. If the period is less than 19 seconds, and cannot be lengthened sufficiently with R57, capacitors C27 and C30 shall be connected into the circuits as shown on applicable schematic diagram referenced on Drawing 8151460 (see 3.5.2).

4.8.2.4 *Search voltage.* Item 1 shall be connected to J3 of the unit under test and the voltage measured from J3 to ground shall be equal to or more negative than minus 35 volts when the dim electrode of I1 is illuminated and a positive 16 volts plus or minus 1 volt when the bright electrode is illuminated (see 3.5.3).

4.8.2.5 *Oscillator output.* The repeller adjust control shall be rotated to its full clockwise position and P1-11 of the unit under test shall be connected to the vertical input on item 2. The output of item 3 shall be connected to the horizontal input on item 2. Adjust frequency of item 3 to obtain a single circular figure. The frequency indicated by item shall be between 1,000 and 1,400 cycles per second. Voltage output at P1-11 as measured on item 4, shall be between 2 and 5 volts (see 3.5.4).

4.8.2.6 *Phase inverter sensitivity.*

4.8.2.6.1 *Wave form at V3 pin 5.* With vertical input of item 2, and the input of item 4 connected to V3 pin 5, turn sensitivity control slowly clockwise. Voltage shall increase smoothly, as indicated on item 4. Adjust the sensitivity control until the positive half of the sine wave flattens slightly. Voltage as read on item 4 shall be between 7 and 12 volts (see 3.5.5).

4.8.2.6.2 *Wave form at V4 pin 5.* With the vertical input of item 2 and the input of item 4 connected to V4 and pin 5, adjust the sensitivity control until the positive half of the sine wave flattens slightly. Voltage shall be between 7 and 12 volts, as indicated on item 4 (see 3.5.5.2).

4.8.2.7 *Sensitivity test for zero degree phase.*

4.8.2.7.1 *Zero degree phase adjust.* Connect P1-11 to the zero degree phase portion of item 5 to J1. Connect item 4 to J1 and item 6 to J2. Connect item 2, to V4 pin 2. Increase the output of item 5 until sharp positive pulses appear on item 2 indicating that V4 is conducting. Input at J1 shall be less than 10 millivolts (see 3.5.6.1).

4.8.2.7.2 *Wave form at V3 pin 2.* Connect item 2 to V3 pin 2. No pulses shall appear on item 2 as in 4.8.2.7.1. Connect item 2 to AFC output J3. No square wave shall be present. Connect item 1 to J3 and measure a positive voltage (see 3.5.6.2).

4.8.2.7.3 *Wave form at J3.* Increase input to J1 to 45 millivolts. Item 2 shall be used to determine that there is no conductance at J3. No square wave shall appear at J3 (see 3.5.6.2).

4.8.2.8 *Sensitivity test for 180° phase.*

4.8.2.8.1 *180° phase adjust.* Connect P1-11 through 180° phase portion of item 5 to J1. Connect item 2 to V3 pin 2. Increase the output of item 5 until sharp positive pulses appear on item 2 indicating V3 is conducting. Input at J1 shall be less than 10 millivolts (see 3.5.7.1).

4.8.2.8.2 *Wave form at V4 pin 2.* Connect item 2 to V4 pin 2. No pulses shall appear on item 2 as in 4.8.2.7.1. Connect item 2 to J3; no square wave shall be present. Connect item 1 to J3 and measure a negative voltage (see 3.5.7.2).

Source: https://assist.dla.mil -- Downloaded: 20261O-22T06:59Z
Check the source to verify that this is the current version before use.

4.8.2.8.3 *Wave form at J3.* Increase input at J1 to 45 millivolts. Item 2 shall be used to determine that V4 does not conduct, and no square wave shall appear at J3 (see 3.5.7.2).

4.8.3 *Environmental.*

4.8.3.1 *Temperature.* The operating and nonoperating temperature test shall be performed in accordance with the high and low temperature test of Specification MIL-E-5272, procedure II, with the exception that the extremes of operating and nonoperating temperatures to which the unit will be subjected shall be as specified in 3.6.1.

4.8.3.2 *Vibration.* The unit, nonoperating, shall be tested in accordance with the vibration test of Specification MIL-T-945, test 2, except the total excursion shall be 0.0156 inch plus or minus 10 percent. Upon completion of the vibration test, the unit shall conform to the requirements specified in 3.6.2.

4.8.3.3 *Shock.* The unit, nonoperating, shall be tilted up through an angle of 30° or 6 inches, whichever is less, using one edge of the indicator as an axis. The unit shall be allowed to fall back freely to the horizontal position on a 2-inch-thick hardwood surface. The tests shall be repeated, using other practicable edges of the same horizontal face as axes, for a total of 4 drops. The unit shall conform to the requirements as specified in 3.6.3 following the series of shock tests.

4.8.3.4 *Relative humidity.* The unit shall be tested in accordance with the humidity test of Specification MIL-E-5272, procedure III, except that the humidity shall be 95 percent. The unit shall conform to the requirements of 3.6.4 upon conclusion of the humidity test.

4.8.3.5 *Fungus resistance.* The unit shall be tested in accordance with the fungus test of Specification MIL-E-5272, procedure I, except no evidence of fungus growth shall be present upon conclusion of the tests (see 3.6.5).

4.8.3.6 *Salt-laden air.* The unit shall be tested in accordance with the salt-spray test of Standard MIL-STD-202, method 101. The unit shall conform to the requirements of 3.6.6 upon conclusion of the salt-spray test.

4.8.3.7 *Altitude.* The unit shall be tested in accordance with the temperature-altitude test of Specification MIL-T-945. The altitude shall be as specified in 3.6.7 and the unit shall be nonoperating for the 30,000 feet test. Upon conclusion of the altitude test, the unit shall operate satisfactorily (see 3.6.7).

4.9 Preservation, packaging, packing and marking. The preservation, packaging, packing and marking shall be inspected to insure compliance with the provisions set forth in the packaging data sheet for Ordnance Part Number 8151460.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for the required level (see 6.2) shall be in accordance with the packaging data sheet for Ordnance Part Number 8151460.

6. NOTES

6.1 Intended use. The Control, Frequency, Automatic, 8151460, provides an output voltage with which a reflex Klystron is tuned to a frequency determined by the band pass of an external tunable cavity. The control is used in the Nike-Hercules Missile System.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether a preproduction sample is required (see 3.1).
(c) Number of preproduction samples required.
(d) Where the preproduction sample tests shall be performed (see 4.2).
(e) Inspection other than specified (see 4.3).
(f) Selection of applicable levels of preservation, packaging, packing and marking (see 5.1).
(g) Applicable stock number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:59Z
Check the source to verify that this is the current version before use.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—(MI)

Preparing Activity:
Army—(MI)
Project No. 4935-A064Y

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:59Z
Check the source to verify that this is the current version before use.